Question 1: VAR Modelling and Multivariate Dynamics

1.1 Data Transformation and Stationarity

We construct a multivariate dataset comprising five financial time series from 2000 to 2025: S&P 500 log returns (`sp500_logret`), GBP/USD log returns (`gbpusd_logret`), and Brent crude log returns (`brent_logret`) are stationary by construction. The 10-year US Treasury yield is first-differenced (`ust10y_dy`) to achieve stationarity. Corporate OAS (`corp_oas_pct`) is retained in levels: the ADF statistic of -3.235 rejects the unit root at the 10% level, and credit spreads are economically mean-reverting.

1.2 Lag Length Selection

We estimate a VAR model on the five stationary variables and select the optimal lag length using information criteria. AIC and HQ select lag 5, while BIC favours lag 2. Given the large sample ($T = 6{,}531$) and the objective of capturing short-run dynamics, we follow AIC/HQ and estimate a VAR(5).

Table 1: VAR lag length selection criteria. Bold indicates selected lag.

Lag	AIC	BIC	HQ
1	−49.21	−49.12	−49.18
2	−49.28	−49.14	−49.23
3	−49.31	−49.12	−49.24
4	−49.33	−49.10	−49.25
5	**−49.35**	−49.07	**−49.26**

1.3 VAR Estimation and Diagnostics

A VAR(5) model is estimated equation-by-equation via OLS: $y_t = c + A_1 y_{t-1} + \cdots + A_5 y_{t-5} + \varepsilon_t$, where y_t is the 5×1 vector of stationary variables and A_i are 5×5 coefficient matrices.

Equation Fit. R^2 values reflect the nature of each series: `corp_oas_pct` achieves $R^2 = 0.999$, reflecting the high autocorrelation of the level series, while return series show low but expected fit: `sp500_logret` (0.017), `brent_logret` (0.009), `ust10y_dy` (0.008), and `gbpusd_logret` (0.032).

Residual Diagnostics. Contemporaneous residual correlations are significant across equations, notably S&P 500 and OAS (−0.369) and S&P 500 and yields (+0.293). This suggests common latent shocks not fully captured by the VAR dynamics. ARCH effects are present in all equations, consistent with financial volatility clustering.

1.4 Granger Causality Tests

Table 2: Granger causality test results. ** denotes rejection at 5%, * at 10%.

Null Hypothesis (no Granger causality)	$\chi^2(5)$	p-value	Conclusion
Brent returns → S&P 500 returns	11.69	0.039	Reject*
GBP/USD → S&P 500 returns	7.43	0.191	Not reject
OAS spread → S&P 500 returns	10.07	0.073	Not reject
S&P 500 returns → GBP/USD	91.88	0.000	Reject**
10Y yield Δ → OAS spread	35.68	0.000	Reject**

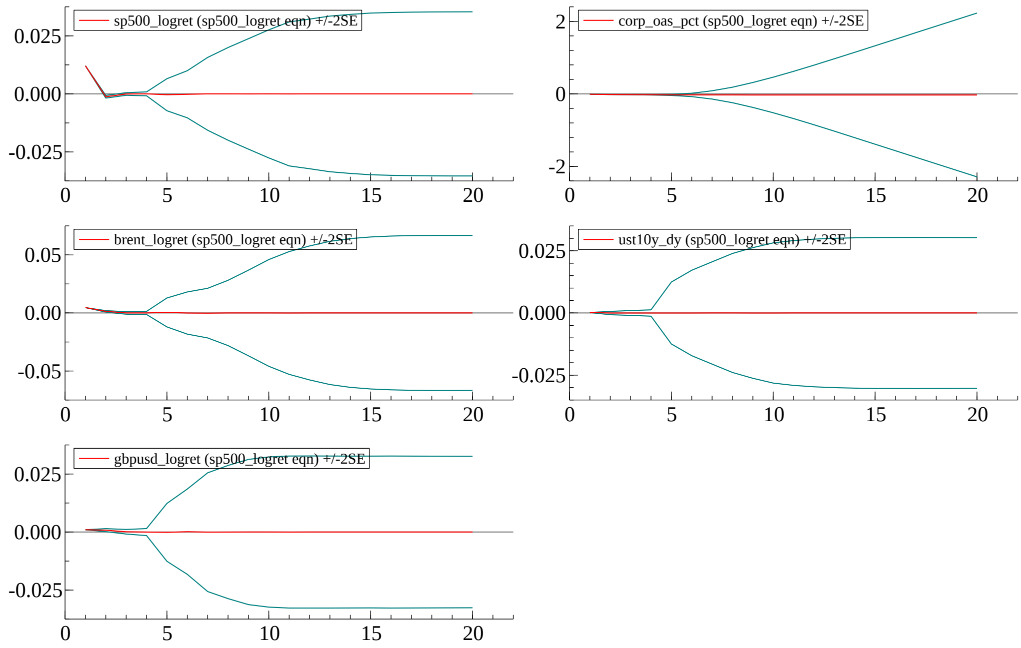
Oil prices Granger-cause equity returns ($p = 0.039$), consistent with cost-push effects. Equity returns strongly lead GBP/USD ($p < 0.001$), reflecting risk-on/risk-off dynamics. Treasury yield changes drive credit spreads ($p < 0.001$), indicating monetary policy transmission. Neither GBP/USD nor OAS spreads provide significant predictive content for equity returns, suggesting equity markets are informationally efficient with respect to these variables.

1.5 Impulse Response Functions and FEVD

We compute orthogonalised IRFs using a Cholesky decomposition ordered as: `sp500_logret`, `corp_oas_pct`, `brent_logret`, `ust10y_dy`, `gbpusd_logret`, assuming equity markets respond fastest to new information.

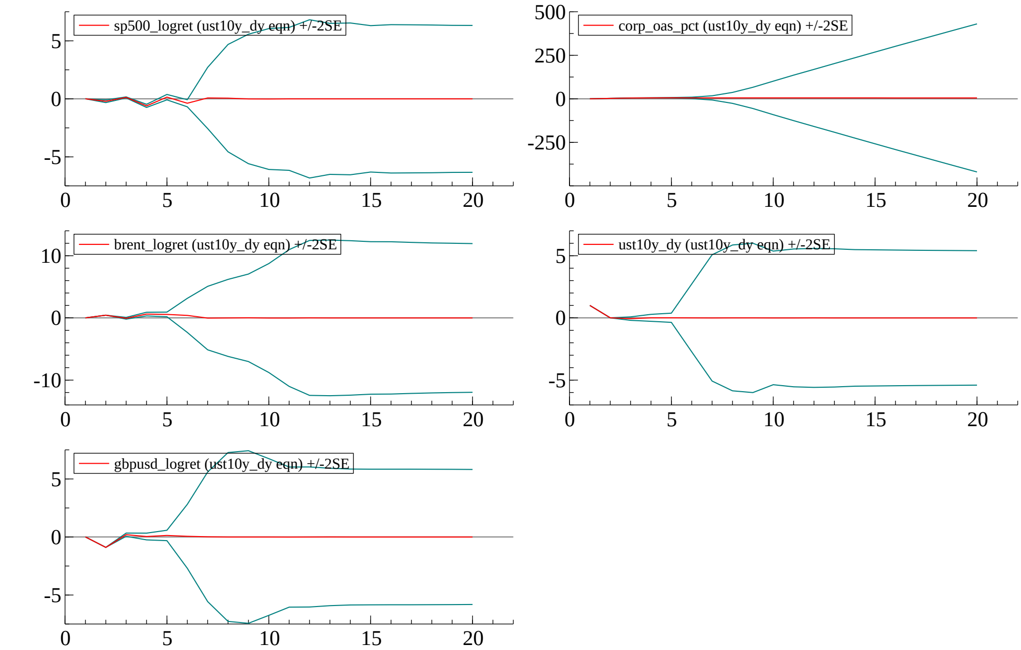
IRF: S&P 500 shock. A positive equity shock triggers a persistent decline in OAS spreads (risk-on compression), modest GBP/USD appreciation, and a small Brent response dissipating within five days.

Figure 1: Impulse responses to a S&P 500 shock. Dashed lines: ±2SE. Horizon in trading days.



IRF: 10-year yield shock. A positive yield shock widens OAS spreads gradually, depreciates GBP/USD as higher US rates attract dollar flows, and produces a small negative but quickly reverting equity response.

Figure 2: Impulse responses to a 10-year yield shock. Dashed lines: ±2SE. Horizon in trading days.



FEVD (20-step ahead): S&P 500 variance is almost entirely self-driven (99.4%). OAS spread variance shifts notably: self-shocks dominate at horizon 1 (86.4%), but S&P 500 contributions rise to 34.6% by horizon 20. Yield change variance remains predominantly self-explained (86.4%). GBP/USD variance is largely self-driven (91.4%), with equities contributing 4.8%.

Professional Communication

Our VAR(5) model reveals key short-run linkages: equity returns strongly lead GBP/USD, with a one-standard-deviation S&P 500 shock appreciating sterling by 0.1% over 5 days. Oil price shocks Granger-cause equities, consistent with cost-push effects. Treasury yield changes Granger-cause credit spreads, though equity shocks account for a larger share of OAS forecast variance (35%) than yield shocks (<1%) at longer horizons, suggesting equity sentiment dominates credit spread dynamics. GBP/USD variance remains largely self-driven (91%), with equities contributing ∼5%. These dynamics imply that trading desks should monitor equity moves for both FX exposure and credit positioning.

Question 2: VECM and Multivariate Cointegration

Data: `market data 2000 2025 CLOSE.csv` (daily). Sample for cointegration/VECM: 2019-06-12 to 2025-03-21.

2.1 Unit Root Tests and Integration Orders

Table 3: ADF Test Results.

Series	ADF level	p level	Lags	ADF diff	p diff	Lags diff	Conclusion
Lsp500_close	−0.6531	0.859	9	−12.021	0.000	8	I(1)
Lgbpusd	−2.1541	0.223	0	−13.426	0.000	9	I(1)
Lbrent_spot	−1.8393	0.361	12	−10.134	0.000	12	I(1)
ust10y_yield_pct	−0.7634	0.830	2	−29.710	0.000	1	I(1)
corp_oas_pct	−3.4820	0.009	11	−10.239	0.000	10	I(0)*

*Rejects unit root in levels; retained in VECM for long-run modelling.

Four series used (`Lsp500_close`, `Lgbpusd`, `Lbrent_spot`, and `ust10y_yield_pct`) are I(1): non-stationary in levels but stationary in first differences. `corp_oas_pct` rejects the unit root in levels (I(0)), reflecting its mean-reverting nature; despite this, it is highly persistent and is retained in the VECM to model the long-run equilibrium between equity valuations, rates, and credit spreads.

2.2 Johansen's Methodology

We apply Johansen's trace test to {log(S&P500), 10y UST yield, corporate OAS} under a restricted constant (CIMEAN) specification. The trace test rejects $r = 0$ at 5% (Trace = 37.64 vs. critical = 35.07; $p = 0.025$) and fails to reject $r \leq 1$ (Trace = 9.02; $p = 0.736$), giving rank $r = 1$. Financial justification: equity valuations, long-term discount rates, and credit risk premia are jointly linked through discounting and risk premia channels.

Table 4: Johansen Trace Test.

H_0	Trace statistic	Critical value (5%)
$r \leq 0$	37.64	35.07
$r \leq 1$	9.02	20.16
$r \leq 2$	3.52	9.14

Reject $r = 0$; do not reject $r \leq 1$. Rank = 1.

2.3 VECM Estimation and Interpretation

The VECM of rank $r = 1$ takes the form $\Delta y_t = \alpha\beta^\top y_{t-1} + c + \sum_{i=1}^{k-1} \Gamma_i \Delta y_{t-i} + \varepsilon_t$. The estimated cointegrating relation (normalised on `corp_oas_pct`) is:

$$\mathrm{ECT}_{t-1} = 1.71 \cdot \mathrm{Lsp500}_{t-1} - 0.104 \cdot \mathrm{ust10y}_{t-1} + \mathrm{corp\_oas}_{t-1} - 15.2 \tag{1}$$

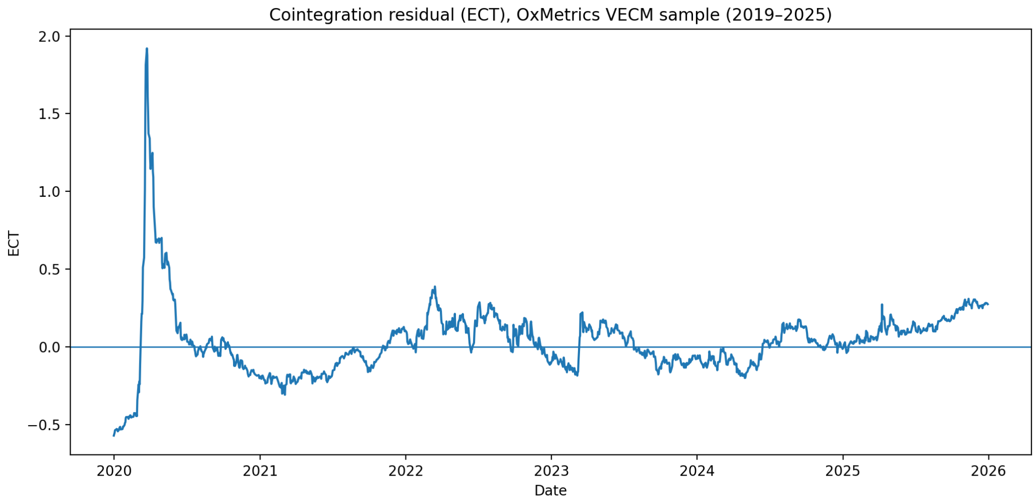
A positive ECT indicates `corp_oas_pct` is high relative to the equilibrium level implied by equity valuations and rates (tighter-than-equilibrium credit conditions).

Table 5: Loading Coefficients $\hat{\alpha}$.

Equation (Δ variable)	$\hat{\alpha}$	t-stat
ΔLsp500_close	0.00321	2.2
Δust10y_yield_pct	0.00931	1.4
Δcorp_oas_pct	−0.01570	−5.3

The adjustment is strongest in Δ`corp_oas_pct` (negative and highly significant), indicating corporate spreads are the main variable correcting long-run deviations. The ECT spikes to its maximum (1.92) on 2020-03-23 during COVID, then gradually mean-reverts as financial conditions normalise.

Figure 3: Error Correction Term ECT_t over time.



2.5 Forecast Comparison: VAR (Differences) vs. VECM

Estimation window: 2010–2019. Hold-out: 2020–2025-03-21. Both models use lag $p = 3$.

Table 6: Out-of-sample accuracy: 1-step ahead.

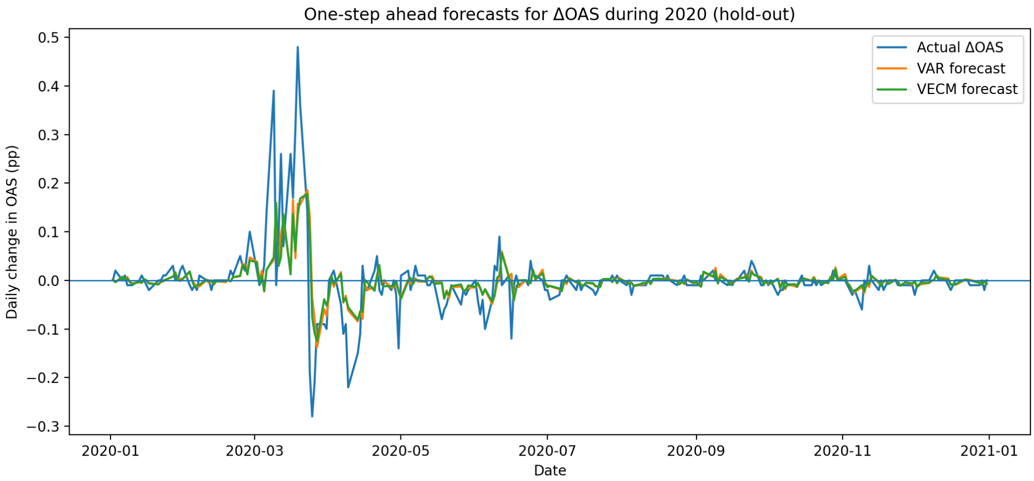
Metric	Series	VAR	VECM
RMSE	$\Delta \log$(S&P500)	0.01331	0.01351
RMSE	Δ10y yield	0.06056	0.06108
RMSE	ΔOAS	0.02783	0.02756
MAE	$\Delta \log$(S&P500)	0.00865	0.00907
MAE	Δ10y yield	0.04599	0.04633
MAE	ΔOAS	0.01375	0.01377

Table 7: Out-of-sample accuracy: 20-day cumulative.

Metric	Series	VAR	VECM
RMSE	$\Delta \log$(S&P500)	0.01329	0.01343
RMSE	Δ10y yield	0.06099	0.06134
RMSE	ΔOAS	0.03303	0.03343
MAE	$\Delta \log$(S&P500)	0.00863	0.00890
MAE	Δ10y yield	0.04624	0.04647
MAE	ΔOAS	0.01425	0.01596

VAR in differences delivers slightly lower RMSE and MAE for most series. The VECM remains useful for interpretation because it explicitly models the long-run equilibrium through ECT_t, even if short-horizon forecasting gains are limited.

Figure 4: Forecast comparison plot.



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Cointegration means key market variables can drift apart in the short run but remain tied by a stable long-run equilibrium. For long-horizon risk management, large shocks appear as temporary disequilibria rather than permanent breaks. The ECT provides a single interpretable measure of disequilibrium: when ECT is far from zero, at least one variable adjusts back toward equilibrium. The cointegrating vector can serve as a long-run hedge ratio. Monitoring ECT also provides a stress signal. In March 2020, ECT jumped sharply, reflecting sudden spread widening and equity stress, signalling that hedges calibrated to normal conditions may need rebalancing.

Question 3: Multivariate Volatility — MGARCH Models

3.1 DCC-GARCH Specification and Estimation

A DCC-GARCH(1,1) model was estimated for five series: equity (EQ), FX, commodity (CMDTY), government yield (GOVYLD), and corporate spread (CORPSPR). Daily series were filtered through a VAR(1) mean equation; residuals were used to estimate five univariate GARCH(1,1) models; standardised residuals then estimated the DCC(1,1) process, with conditional covariance matrix $H_t = D_t R_t D_t$.

Table 8: DCC-GARCH(1,1) Estimation Results.

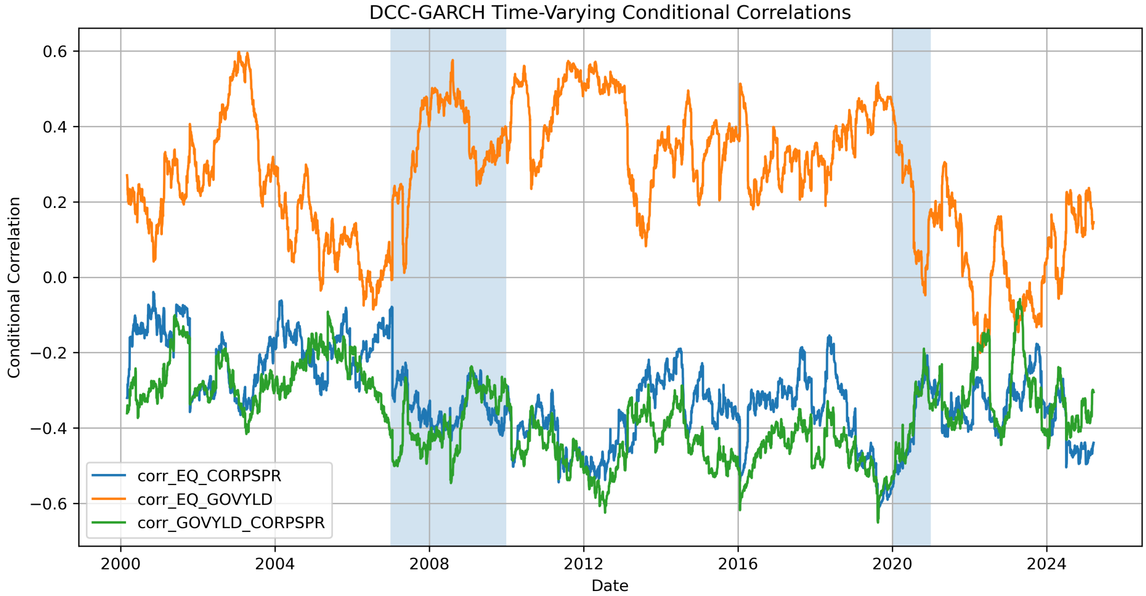
Series	ω	α	β	$\alpha + \beta$	Log-likelihood
EQ	0.0244	0.1187	0.8628	0.9816	−9,005.28
FX	0.0034	0.0528	0.9373	0.9901	−5,141.06
CMDTY	0.0644	0.0869	0.9056	0.9925	−14,165.59
GOVYLD	0.1862	0.0403	0.9542	0.9945	−20,261.66
CORPSPR	0.0519	0.1254	0.8734	0.9988	−11,999.13
DCC	—	0.0100	0.9864	0.9964	+1,660.84

3.2 Trade-off Between Flexibility and Parameter Dimensionality

Volatility persistence is high across all series ($\alpha + \beta$ between 0.9816 and 0.9988), and DCC persistence $a + b = 0.9964$ indicates highly persistent time-varying correlations. For a 5-dimensional system, DCC offers a practical balance: it avoids the $O(N^2)$ parameter explosion of BEKK (75 parameters for $N = 5$) while capturing time-varying correlations through only two additional scalar parameters (a, b) beyond the univariate GARCH models. Unlike CCC, it does not impose constant correlations.

3.3 Time-Varying Conditional Correlations and Crisis Episodes

Figure 5: Time-varying DCC correlations.



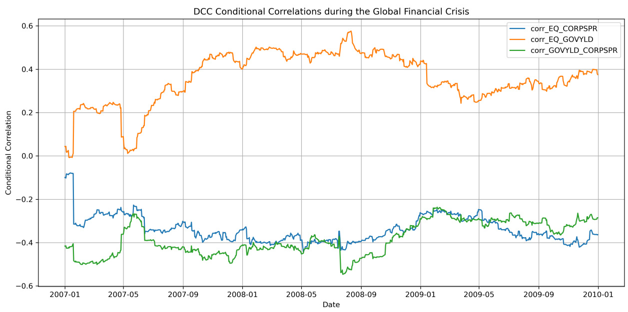
Equity–government yield correlations are mostly positive but unstable, while equity–spread and yield–spread correlations remain largely negative, reflecting typical risk-off behaviour.

Correlation behaviour shifts clearly during stress periods. Table 9 confirms this. The EQ–CORPSPR correlation becomes more negative, falling from −0.319 in the full sample to −0.339 in the GFC and −0.397 during COVID. EQ–GOVYLD rises in the GFC but weakens in COVID, while GOVYLD–CORPSPR remains strongly negative in both crises, indicating stronger stress transmission and weaker diversification in turbulent periods.

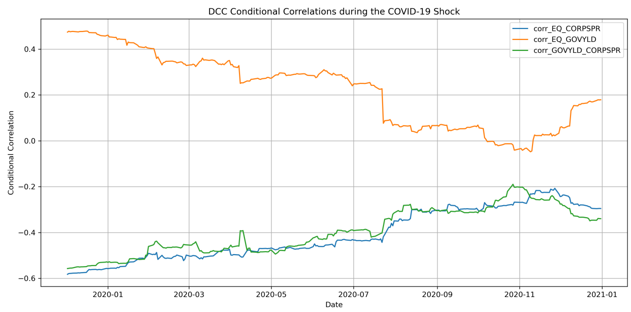
Table 9: Average DCC correlations by period: mean, standard deviation, min and max.

Pair	Period	Mean	Std. Dev.	Min	Max
EQ–CORPSPR	Full Sample	−0.319	0.113	−0.613	0.039
EQ–CORPSPR	GFC	−0.339	0.061	−0.443	0.079
EQ–CORPSPR	COVID	−0.397	0.104	−0.558	0.207
EQ–GOVYLD	Full Sample	0.268	0.178	−0.201	0.598
EQ–GOVYLD	GFC	0.356	0.121	−0.008	0.576
EQ–GOVYLD	COVID	0.202	0.146	−0.048	0.461
GOVYLD–CORPSPR	Full Sample	−0.368	0.109	−0.651	0.058
GOVYLD–CORPSPR	GFC	−0.383	0.076	−0.546	0.237
GOVYLD–CORPSPR	COVID	−0.384	0.093	−0.536	0.190

Figure 6: DCC correlation dynamics during crisis periods.



(a) DCC correlations during the GFC.



(b) DCC correlations during COVID-19.

Table 10: CCC vs. DCC average correlations. CCC estimates capture average dependence but cannot reflect crisis-driven variation.

Pair	CCC	DCC mean (full sample)
EQ–CORPSPR	−0.321	−0.319
EQ–GOVYLD	0.270	0.268
GOVYLD–CORPSPR	−0.361	−0.368

CCC is more parsimonious; DCC is more informative for monitoring correlation shifts.

3.4 Multivariate Risk Metric: Portfolio VaR

Portfolio variance is $w' H_t w$ for equal weights w, and 1-day VaR is $\mathrm{VaR}_t = z_\alpha \sigma_t$ at the 95% and 99% levels. Both models show clear time variation in risk, with pronounced spikes during the GFC and COVID-19. CCC occasionally produces higher VaR estimates during extreme events, suggesting constant correlations may lead to over- or under-estimation when correlations shift rapidly.

Figure 7: Portfolio volatility under DCC and CCC.

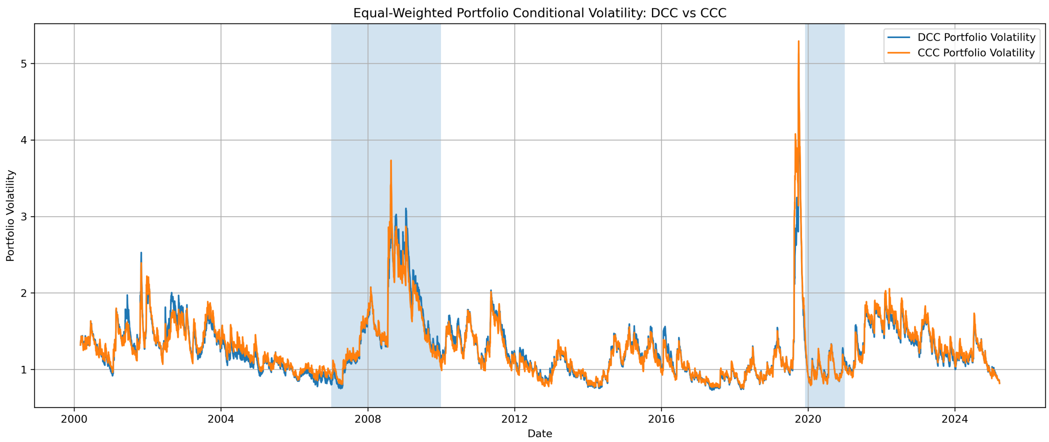
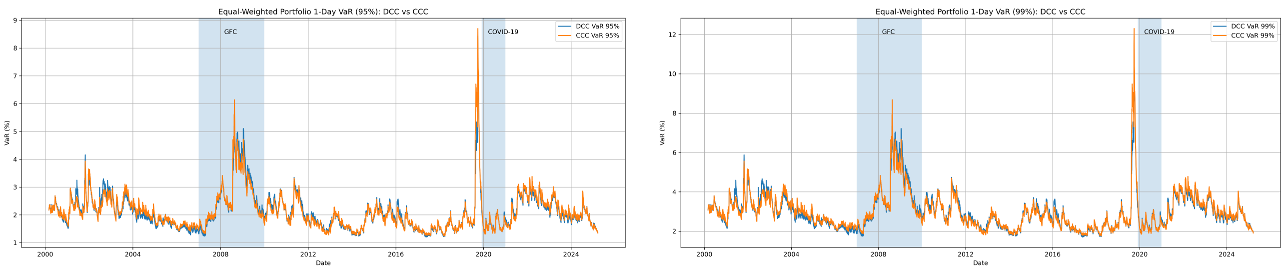


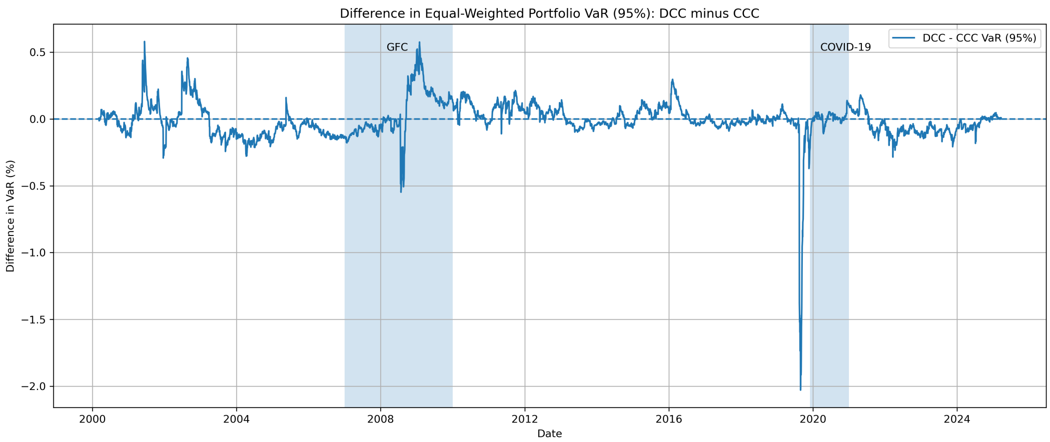
Figure 8: 1-day portfolio VaR under DCC and CCC.



(a) VaR at 95% confidence level.

(b) VaR at 99% confidence level.

Figure 9: Difference between DCC and CCC portfolio VaR at the 95% level.



3.5 Residual Diagnostics

Residual diagnostics indicate the MGARCH models capture much of the volatility structure, but some misspecification remains. Commodity and government yield residuals pass all tests, while equity and corporate spread residuals still show serial correlation, and FX retains remaining ARCH effects. DCC and CCC produce the same diagnostic summary, so the main benefit of DCC remains its richer correlation dynamics rather than better residual fit.

Professional Communication

The DCC-GARCH results indicate that market dependence strengthens materially during stress periods. The equity–corporate spread correlation fell from -0.319 in the full sample to -0.339 during the GFC and -0.397 during COVID, signalling tighter equity-credit stress transmission. The equity–government yield correlation rose to 0.356 in the GFC but declined to 0.202 during COVID, showing crisis-specific behaviour. Portfolio VaR increased sharply under both DCC and CCC, with the largest spikes observed in 2008–2009 and 2020. Residual diagnostics show that only 2 of 5 series pass all adequacy tests, so while the model captures the main systemic-risk patterns, some residual misspecification remains.

Question 4: High-Frequency Data and Realised Measures

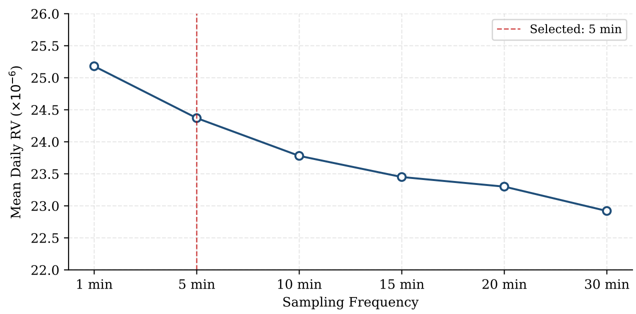
4.1 Asset, Data and Sampling Frequency

We use GBP/USD spot (24/5 FX market) over **2 January–10 October 2025** (202 trading days) at six frequencies: 1, 5, 10, 15, 20, and 30 minutes. All timestamps are UTC; a trading day is defined as 00:00–23:55 UTC Monday to Friday, consistent with the continuous-session convention standard in FX realised variance work (Andersen et al., 2003). We cleaned the data through: (i) weekend removal; (ii) incomplete sessions ($<$ 200 five-minute bars, removing only 1 January 2025); (iii) stale quote *flagging*: 16.9% of overnight bars have identical consecutive closes, retained to avoid distorting the quadratic variation sum; (iv) outlier *flagging*: 143 bars exceed $5\hat{\sigma}$ (14.6 bps/5 min), all at scheduled macro releases (NFP, FOMC, BoE), confirmed as genuine jumps and retained without winsorisation. Intraday log returns $r_{t,i} = \log P_{t,i} - \log P_{t,i-1}$ give daily realised variance $\mathrm{RV}_t = \sum_i r_{t,i}^2$, a consistent estimator of quadratic variation, and of integrated variance absent jumps (Andersen et al., 2003). The **volatility signature plot** (Table 11, Figure 10) shows mean RV rises monotonically from 22.92×10^{-6} (30 min) to 25.18×10^{-6} (1 min). This is a 10% increase, which reflects microstructure noise at fine frequencies (Hansen and Lunde, 2006). The 5-minute series lies in the stable region and is adopted as the baseline (Barndorff-Nielsen and Shephard, 2004) so noise robustness is achieved via frequency selection, not the estimator.

Table 11: Volatility signature, GBP/USD Jan–Oct 2025.

Frequency	Mean RV ($\times10^{-6}$)	Ann. vol (%)
1 min	25.18	7.97
5 min	24.37	7.84
10 min	23.78	7.74
15 min	23.45	7.69
20 min	23.30	7.66
30 min	22.92	7.60

Figure 10: Volatility signature plot. Dashed line: selected 5-minute frequency.



4.2 Realised Volatility and Bipower Variation

Mean annualised RV is **7.46%** (median 6.79%, max 17.49% on 20 Jan 2025, Inauguration Day). Since RV converges to integrated variance *plus* squared jumps (Barndorff-Nielsen and Shephard, 2004), raw

RV overstates persistent diffusive volatility on jump days. We construct **bipower variation** (BV) as the jump-robust alternative: $\mathrm{BV}_t = \mu_1^{-2} \sum_{i=2}^{n} |r_{t,i}||r_{t,i-1}|$, where $\mu_1 = \sqrt{2/\pi} \approx 0.7979$, which converges to integrated variance even with jumps (Barndorff-Nielsen and Shephard, 2004). The jump component is $J_t = \max(\mathrm{RV}_t - \mathrm{BV}_t, 0)$.

Mean BV is 11.5% below mean RV. Separately, the average daily jump share (J_t/RV_t) is **8.4%**, indicating that on a typical day most quadratic variation is diffusive rather than jump-driven. On 10 January 2025 (NFP), the jump share reached 58.4%: BV provides a jump-robust estimate of the continuous component (48.97×10^{-6}) under the maintained assumptions, whereas raw RV (117.80×10^{-6}) would imply a near-threefold rise in baseline volatility. RV and BV are highly correlated ($\rho = 0.929$).

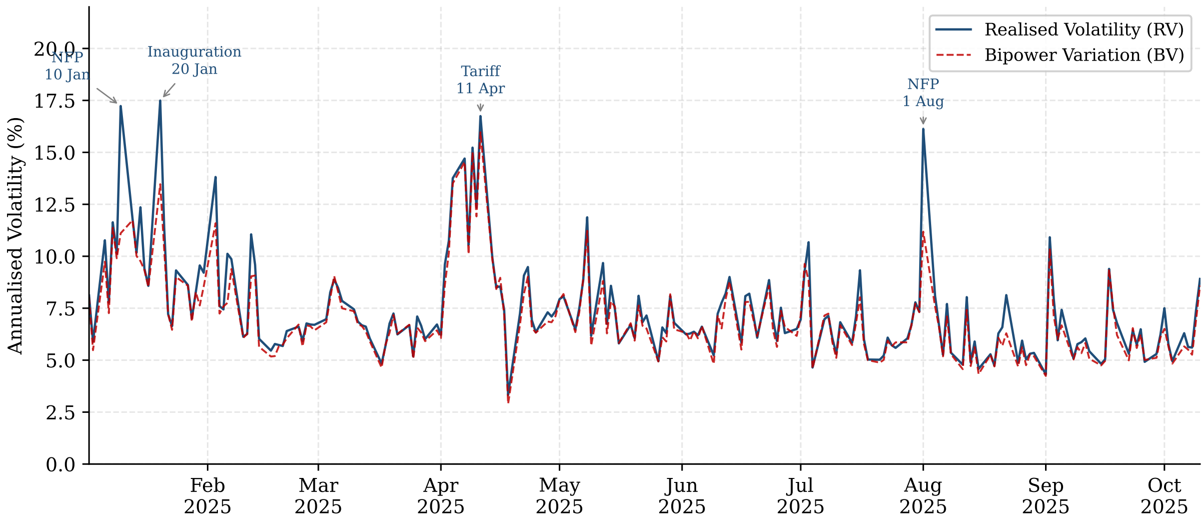
Table 12: RV, BV and jump component ($\times 10^6$).

	Mean	Std	Max
RV	24.37	18.73	121.43
BV	21.58	14.23	101.17
J	2.89	7.57	68.83

Table 13: Top 5 jump days ($\times 10^6$, ann. vol %).

Date	RV	BV	Ratio	Vol
10 Jan	117.80	48.97	58.4%	17.2
01 Aug	103.19	49.65	51.9%	16.1
06 Feb	40.63	23.87	41.2%	10.1
20 Jan	121.43	71.98	40.7%	17.5
22 Aug	26.23	15.72	40.1%	8.1

Figure 11: Annualised RV and BV, GBP/USD 5-minute, Jan–Oct 2025.



4.3 GARCH vs Realised Measures: Crisis Episodes

We estimate GARCH(1,1)-t on daily returns as a model-driven benchmark: $h_t = \omega + \alpha\varepsilon_{t-1}^2 + \beta h_{t-1}$, by QML in G@RCH (OxMetrics), over 2 January–15 August 2025 (162 obs, 40 reserved for hold out).

Table 14: GARCH(1,1)-t estimates. Robust SEs in parentheses.

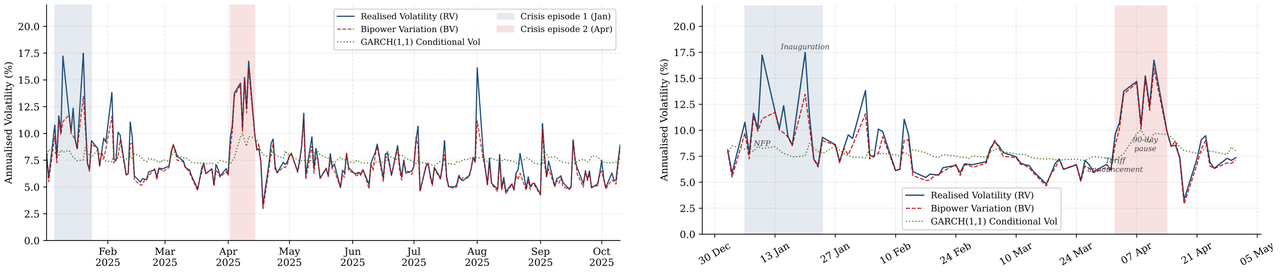
	μ	ω	α	β	ν	$\alpha + \beta$
Estimate	5.46×10^{-4}	5.17×10^{-6}	0.0559	0.7233	20.76	0.779
Std Err	(3.95×10^{-4})	(1.51×10^{-5})	(0.084)	(0.696)	(25.6)	—

Log-likelihood: 631.277. No. obs: 162.

The persistence $\hat{\alpha} + \hat{\beta} = 0.779$ is below the 0.97–0.99 typical for FX in longer samples; neither parameter is individually significant, reflecting the short 9-month window. This produces a notably flat conditional variance path (Figures 12a and 12b).

Episode 1 (January 2025): RV spiked to 121.43×10^{-6} on 20 January (inauguration, 17.49% annualised) and 58.4% of 10 January RV was a jump (NFP). GARCH underestimated peak volatility due to its backward-looking decay where BV correctly identified the spike as transient. **Episode 2 (April 2025):** The US tariff announcement on 2 April produced a jump-share of 19.6% (RV = 36.69×10^{-6}, 9.6% annualised). The subsequent volatility cluster around 7–11 April was primarily diffusive, with RV reaching 111.28×10^{-6} (16.7% annualised) on 11 April but a jump share of only 9.1%, consistent with sustained elevated uncertainty. GARCH lagged throughout.

Figure 12: GARCH conditional volatility vs RV and BV, GBP/USD.



(a) Full sample. Shaded: Episodes 1 & 2.

(b) Zoomed: Jan–Apr 2025.

4.4 HAR-RV Model and Forecast Evaluation

Following Corsi (2009): $\mathrm{RV}_{t+1} = \beta_0 + \beta_d \mathrm{RV}_t + \beta_w \overline{\mathrm{RV}}_t^{(w)} + \beta_m \overline{\mathrm{RV}}_t^{(m)} + u_{t+1}$, where the weekly and monthly components are 5- and 22-day rolling means. Estimated by OLS with Newey–West HAC standard errors over 31 January–14 August 2025 (140 obs).

Table 15: HAR-RV estimates. *, ** denote 5%, 1%.

	Estimate	HACSE
β_0	1.023×10^{-5}	(4.09×10^{-6})**
β_d	0.2937	(0.1415)*
β_w	0.3552	(0.2685)
β_m	−0.0793	(0.2037)
R^2 / Adj. R^2	0.238 / 0.221	

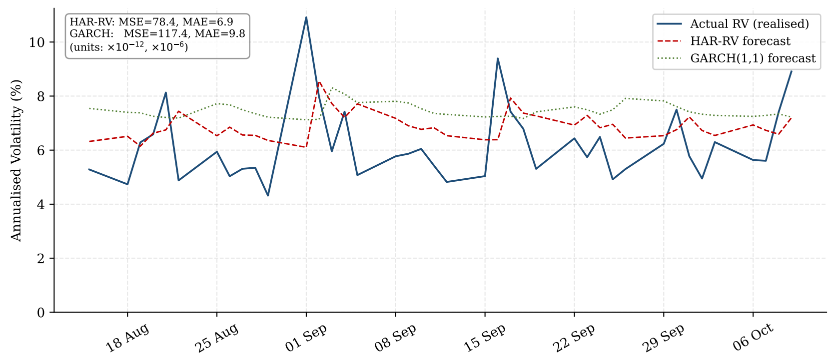
Table 16: Forecast accuracy, hold-out 15 Aug–9 Oct 2025 (40 days).

Model	MSE ($\times 10^{-12}$)	MAE ($\times 10^{-6}$)
HAR-RV	78.40	6.90
GARCH(1,1)-t	117.42	9.79

HAR-RV: 33% lower MSE, 30% lower MAE.
Chow $F(40, 136) = 0.324$ ($p = 1.00$): stable.

Under HAC standard errors, β_d remains significant ($t = 2.08$, $p = 0.040$) while β_w and β_m are not, consistent with Corsi (2009) in short samples. The overall model is highly significant ($F(3, 136) = 14.16$, $p < 0.001$). Diagnostics confirm no serial correlation (AR 1-2: $F = 1.98$, $p = 0.14$) and no ARCH effects ($F = 1.63$, $p = 0.20$). HAR-RV achieves **33% lower MSE** and **30% lower MAE** than GARCH over the 40 day hold out, with stable coefficients (Chow test $p = 1.00$). The improvement reflects RV's informational advantage: it directly captures intraday variation on the day of a shock rather than updating via a single lagged squared return.

Figure 13: One-step-ahead forecasts vs actual RV, hold-out sample (40 days).



To validate the choice of 5-minute sampling, Table 17 reports HAR-RV forecast performance across all six frequencies. Under HACSE, β_d is significant at 5% for 5-, 10-, 20- and 30-minute frequencies but not for 1- or 15-minute, while β_w is insignificant throughout. Forecast MSE deteriorates broadly as sampling coarsens beyond 5 minutes, with 30-minute performing 116% worse. The 1-minute HAR produces the lowest MSE but shows a significant RESET test ($p = 0.028$) and borderline ARCH effects ($p = 0.078$), both consistent with microstructure noise distortion. The 5-minute frequency avoids these problems and remains the preferred baseline.

Table 17: HAR-RV forecast performance by sampling frequency, hold-out 15 Aug–9 Oct 2025. HACSE throughout; *, ** denote 5%, 1%. † RESET test significant ($p < 0.05$).

Freq	β_d	t_d	β_w	t_w	R^2	MSE ($\times 10^{-12}$)	MAE ($\times 10^{-6}$)
1-min†	0.285	1.83	0.438	1.53	0.302	50.88	5.97
5-min	**0.294**	**2.08***	**0.355**	**1.32**	**0.238**	**78.40**	**6.90**
10-min	0.322	2.29*	0.295	1.26	0.221	92.61	7.67
15-min†	0.251	1.82	0.283	1.39	0.140	120.74	8.46
20-min	0.304	2.15*	0.296	1.31	0.207	113.00	7.58
30-min	0.304	2.38*	0.234	1.31	0.165	169.49	8.96

Bold: selected baseline. All Chow tests $p = 1.00$.

4.5 Practical Challenges

Microstructure noise: Sampling too finely inflates RV via bid–ask bounce and discreteness as seen in Table 11. For GBP/USD, the 10% rise in mean RV from 30-minute to 1-minute sampling is moderate, reflecting the pair's high liquidity; for less liquid instruments the distortion would be far more severe. Sampling at 5 minutes, where the signature plot stabilises, is the standard remedy. **Jumps:** Discontinuities inflate RV above the persistent diffusive component. BV isolates this but is not noise-robust, reinforcing the need for both frequency selection and a jump-robust estimator in tandem. **Asynchronous trading:** Non-synchronous quoting induces the Epps effect (downward-biased covariances at fine frequencies), requiring refresh-time or previous-tick interpolation in any cross-asset extension. **Inference:** Jumps, noise, and the U-shaped intraday FX periodicity all complicate finite-sample inference for RV and motivate robust methods. Confidence intervals should use realised kernels and HAR standard errors require Newey–West HAC corrections for overlapping rolling averages, as applied here.

Professional Communication

Using five-minute GBP/USD data over January–October 2025, realised variance materially outperforms GARCH. HAR-RV achieves 33% lower MSE and 30% lower MAE over a 40-day hold-out, with stable coefficients confirmed by a Chow test. The advantage is immediacy: RV captures elevated intraday variation on the day of a macro shock, whereas GARCH requires several days to respond. Bipower variation further separates transient jumps, up to 58% of daily variance on NFP days, from the persistent component relevant to risk budgeting. Adopting a HAR-RV framework could improve volatility inputs for VaR and reduce the risk of overstating baseline volatility following large but one-off events.

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